Course Introduction

CO2003 — Data Structures and Algorithms

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Faculty of Computer Science and Engineering
Ho Chi Minh City University of Technology

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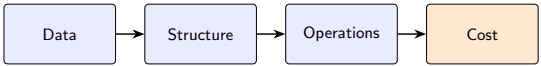
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Why Data Structures and Algorithms?



- Programs store and transform data.
- A structure organises the data.
- An algorithm performs the work.
- We also care about **cost**.

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A Compact Slogan

Niklaus Wirth

$$\text{Program} = \text{Data structures} + \text{Algorithms}$$

- The rest of the course fills in both sides.
- Abstract data types vs. C++ details come in the next deck.

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Course Learning Outcomes

- **L.O.1** Determine the complexity of algorithms
- **L.O.2** Use data structures covered in the course
- **L.O.3** Select appropriate sorting and searching algorithms for different problem contexts.

Source: DCMH.CO2003.14.1

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L.O.1 and L.O.2

- **L.O.1** Determine the complexity of algorithms
  - **L.O.1.1** Explain asymptotic notations such as Big-O, Big-Θ, and Big-Ω.
  - **L.O.1.2** Determine the time complexity of algorithms involving iterative and recursive structures.
- **L.O.2** Use data structures covered in the course
  - **L.O.2.1** Explain the organization and basic operations of data structures.
  - **L.O.2.2** Use data structures to perform basic operations in problem solving.
  - **L.O.2.3** Implement data structures and their specific operations using programming code.

Source: DCMH.CO2003.14.1

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L.O.3

- **L.O.3** Select appropriate sorting and searching algorithms for different problem contexts.
  - **L.O.3.1** Explain the processing steps of searching algorithms (on the data structures covered in the course) and sorting algorithms (including topo-sort).
  - **L.O.3.2** Implement algorithmic procedures using programming code and illustrative diagrams.
  - **L.O.3.3** Select appropriate sorting and searching algorithms based on problem criteria.

Source: DCMH.CO2003.14.1

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Lecture map

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Lecture Map (Weeks 1–8)

15 lecture weeks + midterm (week 8).

Week	Theory
1	Intro + programming + complexity
2	List (array list, singly linked)
3	List (doubly linked)
4	Stack + queue
5	Sorting
6	Sorting + tree concepts
7	Binary search tree
8	**Midterm** (no lecture)

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Lecture Map (Weeks 9–16)

Week	Theory
9	Advanced tree (AVL)
10	Advanced tree (Splay + B-tree)
11	Heap + heap sort
12	Search + hash
13	Hash + graph
14	Graph
15	Graph (advanced topics)
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Labs (planned)

Six lab sessions. Topics below are **planned** and may shift.

Lab	Planned topics
1	OOP + recursion + array / singly / doubly lists
2	Search (linear + binary) + stack/queue + sort
3	In-lab test + BST + AVL
4	Splay tree + hash
5	Heap + graph
6	In-lab test + graph

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Assignments (planned)

Three course projects. Scope and windows are **planned**.

	Planned scope	**Window**
A1	List (array / doubly) + stack/queue + sort + application	Weeks 2–7
A2	BST + AVL + heap + application	Weeks 5–11
A3	Hash + graph + application	Weeks 9–15

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Implementation Language

- Executable code in this course is **C++**.
- Today we review pointers, basic OOP, and recursion — required before lists and trees.
- Complexity is analysed from C++ fragments or a clear algorithm description.

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Textbooks

Required

- Drozdek, *Data Structures and Algorithms in C++* (4th ed.)
- Nguyen Trung Truc, *Cau truc du lieu va giai thuat*

References

- Cormen et al., *Introduction to Algorithms*
- Shaffer, *Data Structures and Algorithm Analysis in C++*

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Today's Three Decks

1. Course introduction (this deck)
2. Programming review: pointers, OOP, recursion
3. Complexity: loops *and* recurrences

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- CO2003 pairs structures with algorithms and their cost.
- Official CLOs cover complexity, data structures, and sort/search.
- Fifteen lecture weeks plus a midterm week; six planned labs; three planned assignments.
- C++ is the implementation language.

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